Subject Code: 25ES11D/21D

Subject Name: Introduction to Mechanical Engineering

Module - 1

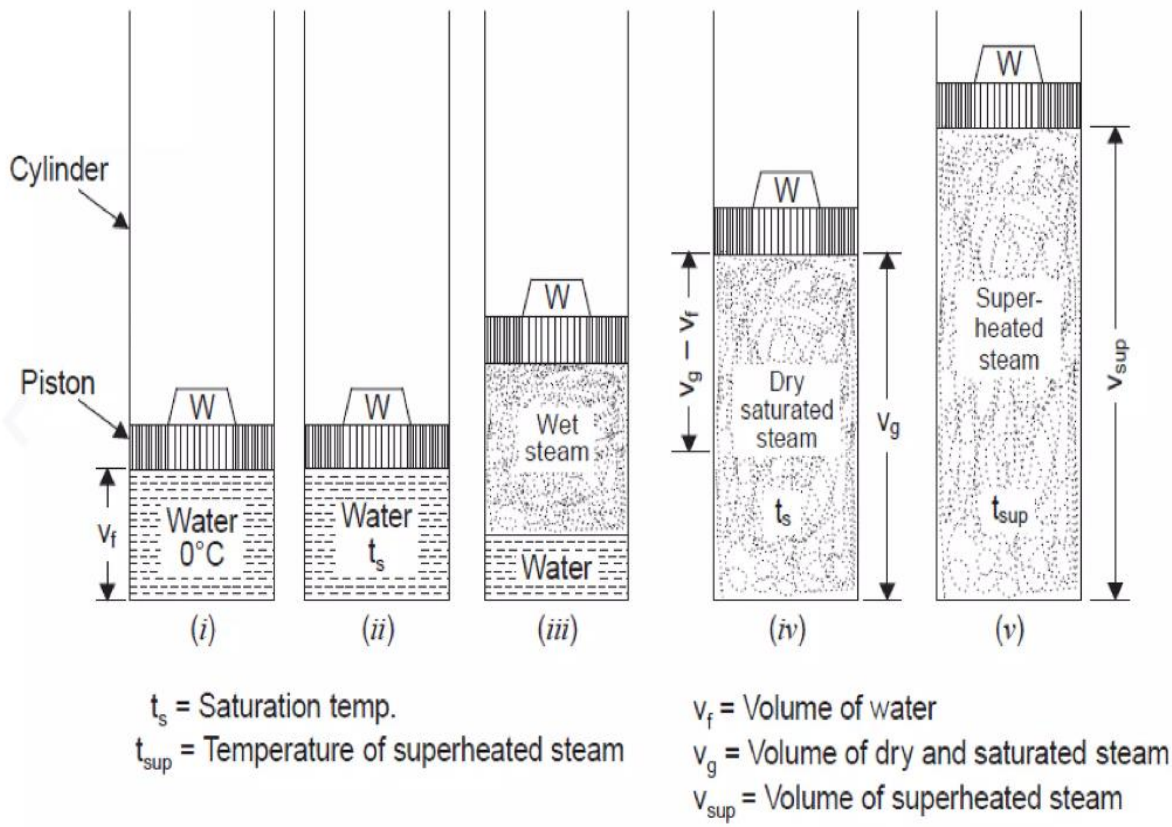
Steam Formation: Steam formation and its representation on T-h diagram, conceptual approach on Properties of steam – Specific Volume, Enthalpy, External Work of Evaporation and Internal Energy.

Energy, it's Sources and Power Plants: Review of energy sources; Construction and working of hydel power plant, solar thermal power plant, nuclear power plant and wind power plant.

Refrigeration and air conditioning: Refrigeration effect, working principle of Vapor Compression Refrigeration (VCR) and Vapor Absorption Refrigeration (VAR) systems, difference between VCR and VAR, ton of refrigeration, COP, refrigerants and their properties. Working Principles of room air conditioning and applications of air conditioners.

Steam: Steam is an invisible gas created by adding heat energy to water. It is liquid water changed to its gaseous state.

Steam formation with T- H Diagram:



Consider 1 kg water in friction free cylinder. Let piston applies a constant pressure on water as shown in figure. Assume that this experiment is carried out at constant atmospheric pressure. The water is heated from the other end of the cylinder.

State 1 (Subcooled Liquid): Consider liquid Water is heated from say 0 ^{0}C temperature, at this state the water is assumed in liquid state and below the boiling (Saturation) temperature. Hence it is called as subcooled liquid state. Water remains in subcooled state till it reaches the saturation temperature. At subcooled state, upon adding heat, there is rise in temperature of water.

State 2 (Saturated Liquid): When water temperature reaches to saturation temperature , 100 ^{0}C, the water starts boiling. The further addition of heat will no rise the temperature of the water. From the state 2, the liquid water starts converting to vapour state. .At this point dryness fraction will be x = 0.

State 3 (Wet Steam): At the state 3, water is in two state i.e liquid and vapour state. Since the vapour is in contact with water, steam is called as wet steam. Further addition of heat there is no rise in saturation temperature and it remains in wet steam till all liquid is converted into vapour state. At the wet state condition, the dryness fraction (x) varies from 0 to 1.

State 4 (Dry Steam): When all liquid water converted into vapour state, vapour becomes dry steam. At this point dryness fraction will be x = 1. Further addition of heat will increase the temperature of the steam.

State 5 (Superheated Steam): On further addition of heat, the temperature of steam starts increasing and the steam is super heated steam.

Temperature v/s Enthalpy Diagram:

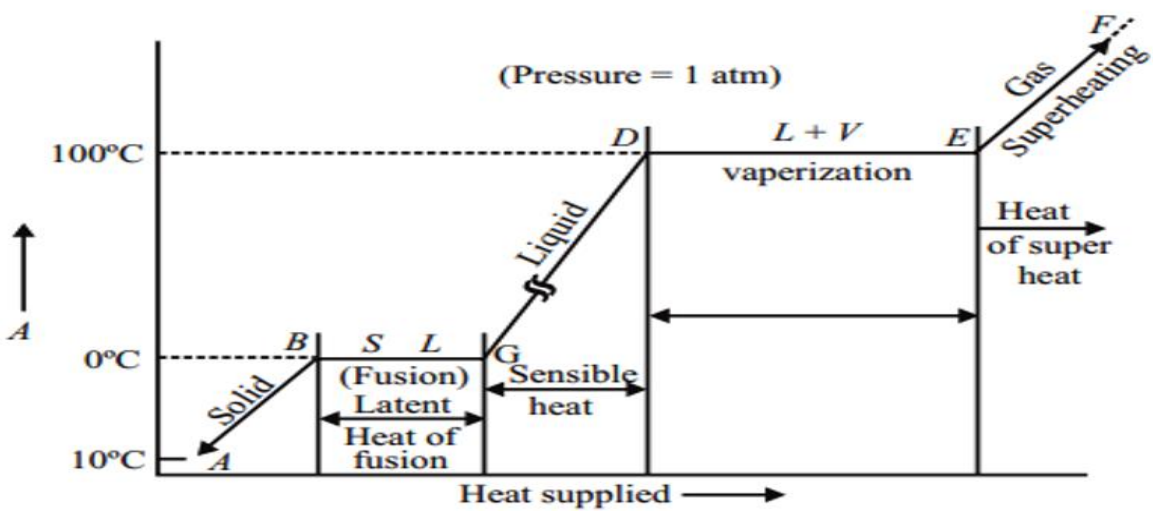


Figure: Temperature v/s Enthalpy diagram

Properties of Steam / Some definition on steam:

- **Enthalpy or Sensible or Total Heat of Water[h_f] :** Amount of heat required to convert 1 kg of water from ice point to boiling point at constant pressure. It is represented by, h_f.

$$h_f = 4.18[T_{sat} - T_{ice}] \ldots\ldots kJ/Kg.$$

- **Latent heat of Vaporization[h_{fg}]:** Amount of heat required to convert 1 kg of water to completely steam. It is represented by, h_{fg}.

For water, $h_{fg} = 2260$ kJ/Kg.

- **Enthalpy of wet steam [h_w]:** Amount of heat required to convert 1 kg of water from ice point to given dryness fraction. It is represented by, h_w.

$$h_w = h_f + x\, h_{fg} \ldots\ldots kJ/Kg.$$

- **Enthalpy of dry steam [h_g]:** Amount of heat required to convert 1 kg of water from ice point to dry steam. It is represented by, h_g.

$$h_g = h_f + h_{fg} \ldots\ldots kJ/Kg.$$

- **Enthalpy of super heated steam[h_{sup}]:** Amount of heat required to convert 1 kg of water from ice point to given degree of superheat of steam. It is represented by, h_{sup}.

$$h_{sup} = h_g + C_{ps}[T_{sup} - T_{sat}] \ldots\ldots kJ/Kg$$

Where, C_{ps} = Specific Heat of Steam = 2.1 kJ/KgK

- **Degree of Superheat:** The difference between super heated temperature and the saturation temperature is called degrees of saturation.

$$[T_{sup} - T_{sat}] = \text{Degree of Superheat.}$$

- **Dryness Fraction(x) or Quality of steam[x]:** It is the ration of Mass of dry steam to mass of wet steam.

$$x = \frac{Mass\ of\ Dry\ Steam}{Mass\ of\ Wet\ Steam} = \frac{m_g}{m_g + m_f}$$

- **Super Cooled Liquid:** Liquid that exist below its boiling temperature.
- **Saturated liquid:** It is state of a substance at which liquid starts vaporizing on heating at constant pressure.
- **Saturated Vapour:** It is state of a substance at which liquid starts condensing on cooling at constant pressure.
- **Saturation Temperature:** The temperature at which a liquid starts boiling at the given pressure is called as saturation temperature.

Ex: $T_{sat} = 100\ ^0C$ at atmospheric pressure (1.01325 bar)

i) Specific Volume (v): Specific volume is the volume of a substance per unit mass, and the inverse of density. It is an intensive property of a substance.

i.e $v = V/m = 1/\rho$

where v - Specific volume; V – Volume of a substance;
m – mass of a substance; ρ – density of a substance

ii) Enthalpy (H): Enthalpy is the heat absorbed or released by a system at constant pressure. Enthalpy is the energy in a thermodynamic system that equals the internal energy plus the product of volume and pressure.

i.e $H = U + P*V$

where, U - Internal energy; P - pressure; V - Volume

- At a constant pressure, heat is the same as enthalpy.

iii) External work of Evaporation (W): The amount of heat absorbed due to the increase in volume when water is converted into steam.

iv) Internal Energy (U): the sum of kinetic and potential energy of molecules in a system

Internal energy indicators

- The energy entering the system is POSITIVE (+), indicating that heat is ABSORBED.
  i.e Work done on the system

- The energy entering the system is NEGATIVE (-), indicating that heat is RELEASED.
  i.e **Work done by the system**

Energy it's Sources and Power Plants:

Review of energy sources discuss on hydel power plant, solar power plant, nuclear power plant and wind power plant with respect to Construction and working.

Hydel Power Plant: A hydroelectric power plant comprises a set of facilities and electromechanical equipment used to transform water's **potential energy** into **electrical energy.**

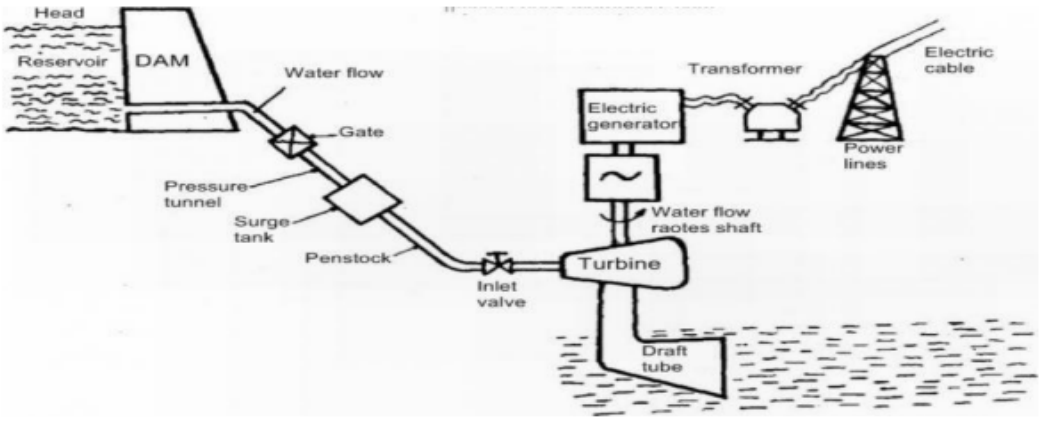


Figure: Hydel power plant

Components of Hydro Power Plant

1. **Reservoir:** It stores large amount of water behind the dam. The energy stored in reservoir is potential energy. The height of the water stored in reservoir will determine the potential energy stored in water.
2. **Surge Tank:** A surge tank is a small reservoir or tank which is open at the top. It is fitted between the reservoir and the turbine inlet. When there is sudden reduction in load on the turbine, the governor closes the gates of the turbine to reduce the water flow. This causes pressure to increase abnormally in the penstock. This is prevented by using a surge tank, in which the water level rises to reduce the pressure. On the other hand, the surge tank provides excess water needed when the gates are suddenly opened to meet the increased load demand.
3. **Penstock:** A penstock is a huge steel pipe which carries water from the reservoir to the turbine. Potential energy of the water is converted into kinetic energy as it flows down through the penstock due to gravity.
4. **Water Turbine:** It converts high kinetic energy into mechanical energy once the water flows over the turbine blades.
5. **Generator:** The generator shaft is coupled to turbine shaft. In generator, mechanical energy is converted into electrical energy. This electric energy is further transmitted to grids.
6. **Draft Tube:** It is pipe through which water discharged into river.

Working of Hydro Power Plant:

Hydroelectric power plant (Hydel plant) utilizes the potential energy of water stored in a dam built across the river. The potential energy of the stored water is converted into kinetic energy by first passing it through the penstock pipe. The kinetic energy of the water is then converted into mechanical energy in a water turbine. The turbine is coupled to the electric generator. The mechanical energy available at the shaft of the turbine is converted into electrical energy by means of the generator.

Advantages:

- No fuel is required as potential energy is stored water is used for electricity generation
- Neat and clean source of energy
- Very small running charges - as water is available free of cost
- Comparatively less maintenance is required and has longer life
- Serves other purposes too, such as irrigation

Disadvantages:

- High construction cost is required due to construction of dam.
- long transmission line is required to transmit this hydroelectric power.
- It doesn't supply constant hydroelectricity due to the availability of water.

Solar Power Plant

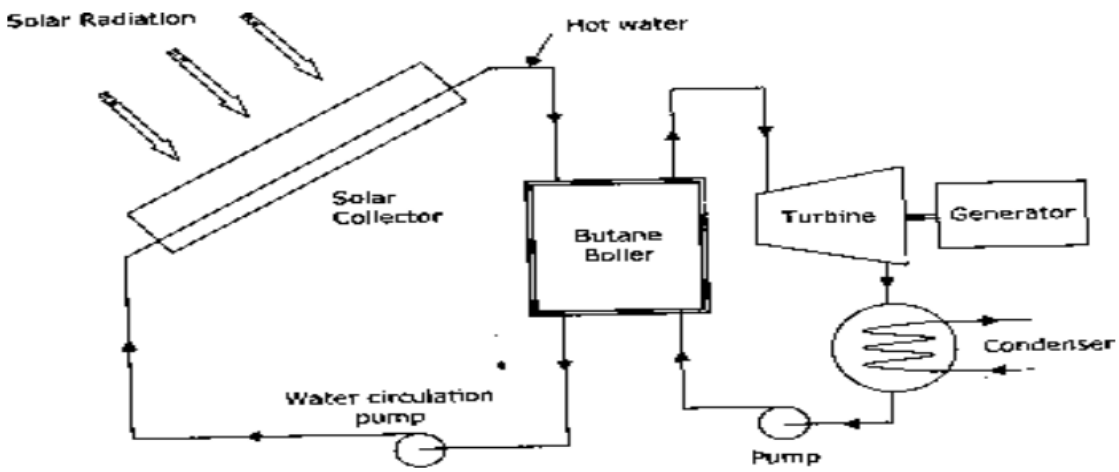


Figure: Schematic Representation of Solar Power Plant

Components of Solar Power Plant

- **Solar Collector:** A solar collector is a heat exchanger, where it transmits the heat from sun radiation to water. After absorption of heat, water density reduces and high density cold water pushes the hot water away from the collector.
- **Heat Exchanger (Butane Boiler):** It is a heat exchanger, where hot water from solar collector will transfer heat to butane. After absorbing heat, butane converts into super heated vapour. Butane is selected as working fluid as it boils at low temperature.
- **Turbine:** Turbine is work producing device. It converts kinetic energy of the butane vapour into mechanical energy when vapour is made to expand on the turbine blades. The expanded vapour is in binary state of vapour and liquid. This expanded steam enters into condenser.
- **Generator:** Generator is a electric device, which converts mechanical energy into electric energy. Thus produced electric energy is transmitted to electric grids.
- **Condenser:** Condenser is a heat exchanger. Here vapour from turbine enters into condenser tubes and loses heat to the cold water. After losing heat, the steam converted into liquid state.
- **Pump:** Pump is a work absorbing device. The liquid butane is fed back to boiler with help of feed pump. It will regulate the flow of butane in heat exchanger so as to generated the vapour at desired quantity and pressure levels.

Working of Solar Power Plant

The solar power plant utilizes the heat energy from the solar radiations to heat liquid water in solar collector. This water, after absorbing heat, enter into heat exchanger. In heat exchanger hot water transmits heat to butane in liquid state. After absorbing heat, the liquid butane turns into vapour at high pressure. This high-pressure vapour is converted into kinetic energy with the help of nozzle. This expanded steam is then passed over the turbine blades. Where, high kinetic energy of steam is converted into mechanical energy. Thus, produced mechanical energy is converted into electrical energy with help of generator. This electric energy transmitted to electric grids. The vapour expanded in turbine is further cooled in condenser. After condensation vapor is converted into liquid butane. This butane is pumped back to heat exchanger with the help of feed pump.

Advantages and limitation

Advantages:

- Endless amounts of energy, free of cost.
- Modern systems work efficiently even in winter
- Solar power is pollution-free and causes no greenhouse gases to be emitted after installation
- Reduced dependence on import of foreign oil and fossil fuels like coal based plant etc.
- Plant can be installed at remote locations. Hence, power transmission losses are low.

Disadvantages:

- Cost of power generation is high.
- Needs thermal power storage system for uninterrupted working during when sun is unavailable.
- Needs very large solar collector area for installation.
- Cannot supply continuous electric power.
- Only suitable where favorable sun-shine conditions are available.
- Low thermal efficiency.

Nuclear Power Plant

Nuclear power plant is a **thermal power plant** in which it makes **use of nuclear fission** process in order to **generate electricity.**

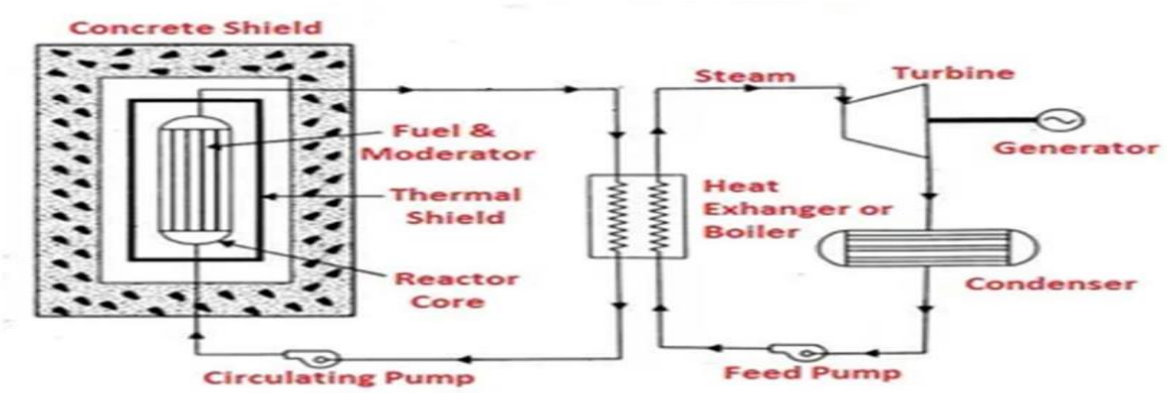


Figure: Schematic Representation of Nuclear Power Plant

Components of Nuclear Power Plant

- **Nuclear Reactor:** In nuclear reactor the **fission reaction** is **initiated by bombarding heavy nucleolus** of the fuel **(Uranium) with the neutron.** This fission reaction generates **enormous amount of heat energy**. This **heat is absorbed by the fluid** and **exchanges heat to secondary fluid** in heat exchangers.
- **Moderator:** It **reduces the kinetic energy** of the **neutron** to **control the chain reaction**. Generally **graphite** is used as moderator.
- **Thermal Shield:** This concrete structure **avoids leakage of radiation** to the **outer atmosphere.**
- **Coolant:** Coolant **absorbs heat** from the **nuclear reactor** that **keeps the reactor cooled.**
- **Heat Exchanger (Boiler):** It is a heat exchanger, absorb heat and converts water into superheated steam.
- **Steam Turbine:** It **converts kinetic energy** of the **steam into mechanical energy** when steam is made to expand on the turbine blades.
- **Generator:** which **converts mechanical energy into electric energy.**
- **Condenser:** Condenser is a heat exchanger, which **convert steam into liquid state.**
- **Pump:** The **liquid water is fed back to boiler** with help of feed water pump.

Working of Nuclear Power Plant:

The **nuclear power plant utilizes** the **heat energy generated during the fission reaction** which produce enormous heat energy. This **heat** is **absorbed by a coolant while passing through the thermal shield**. This heat is **further transmitted secondary liquid (water).** The **boiler to convert liquid water into superheated steam.** The **superheated steam** is made to **pass over the turbine blades** which **generate the rotational energy.** Since the **generate** is coupled with turbine which **convert rotational energy into electrical energy** and thus electrical energy is produced.

The steam comes out of the turbine is passed through the condenser which convert steam into liquid state and further it is feedback to boiler with the help of feed water pump

Advantages and limitation

Advantages:

- Nuclear Power Plant requires less space as compared to other Power plants.
- Well suited for large scale power generation.
- It has very high energy density compared all other power plants.
- Less fuel consumption and no fuel handling compared to thermal power plants.
- Increased reliability of operation.
- Compact and simple in maintenance.

Disadvantages:

- Nuclear Power Plant has a High initial cost.
- The danger of radioactivity hazards is always persisting.
- The disposal of fission products is a big problem.
- The maintenance cost is always higher.
- Working condition is always harmful to the health of the workers.

Wind Power Plant

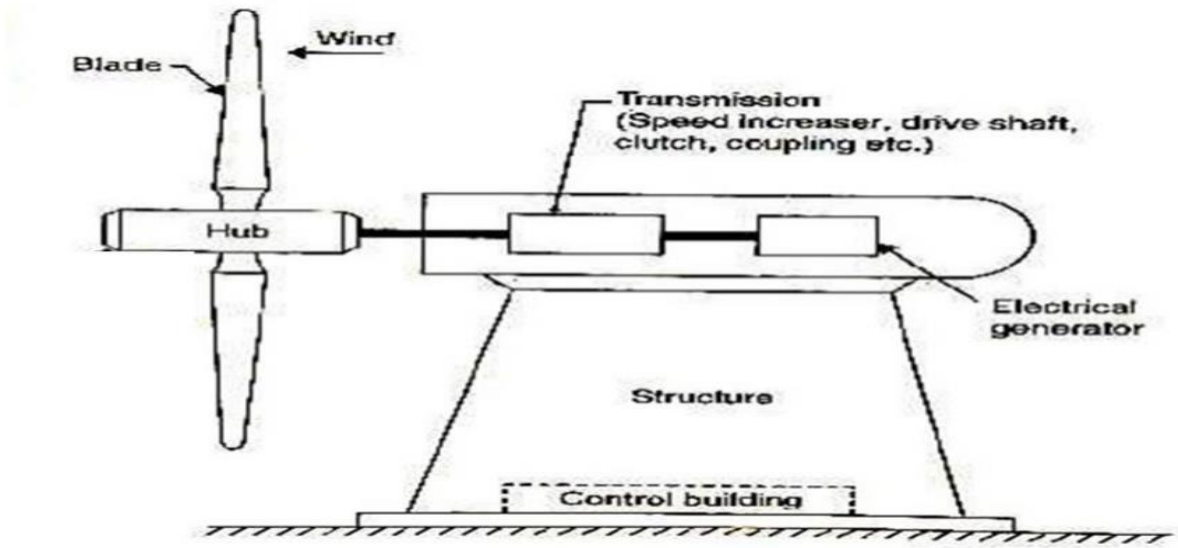


Figure: Horizontal Axis Wind Turbine

Components of Wind Power Plant

- **Turbine Blades:** It is propellers with two, three or five blades mounted on the horizontal shaft (this gives higher output than when they are mounted on the vertical shaft) and made of a lightweight material such as carbon fiber, fiberglass or wood, that is strong enough to resist wind forces. It is built with aero foil structure due to which kinetic energy of the wind is converted into rotary motion of the turbine shaft.
- **Gear Box:** A Mechanical gearbox is used to increase the rotational speed of the generator.
- **Generator:** An Electrical generator is used to produce the electrical power which is coupled on gear shaft.
- **Support Structure:** It supports the rotors, gearbox, generator and axillary equipment.

Working of wind power plant:

The wind power plant utilizes the **kinetic energy** from the **wind** to **mechanical energy** in turn to **produce electrical energy**. **Wind energy is created** when the **sun heats the earth atmosphere unevenly** causing **thermosiphon effect.** This thermosiphon effect is the cause for formation of the wind. Hence, **Wind energy**, which is actually a **secondary component of solar energy.**

When the **wind flows over the turbine blade,** due to its **aero foil shape**, the **kinetic energy of the wind is converted into rotary motion of the shaft.** These **blades are connected to a hub**. Hub is further **connected to gear box.** Generally, the **speed of the rotor is very low**. This isn't **enough to produce the high electric power. To increase the speed of the rotor, a gear box is employed**. Thishigh-speed shaft connects to an electrical **generator that converts the**

mechanical energy from therotation of the blades into electrical energy. Spinning between 11 and 20 times per minute, **each turbine can generate a maximum 1.5 megawatts of electricity enough to power,** on average, morethan 500 residential homes.

Advantages:

- Air as a fuel is free and inexhaustible.
- It is clean source of energy and does note pollute the environment.
- The cost of electricity is too low and wind turbine could be used over more than 20 years
- It's cheap as only the installation and maintenance cost is required.

Disadvantages:

- It takes a lot of research and effort to decide the location where wind power plant has to beinstalled, due to fluctuating pattern of wind.
- They are the greatest disadvantage to local bird population as they die due to collision withblades.
- Noise pollution is the one of the major disadvantage.
- Wind power plant is only useful to the countries with coastal or hilly areas.

Refrigeration and Air conditioning:

Refrigeration: is the removal of unwanted heat from a selected object, substance, or space and its transfer to another object, substance, or space.

OR

Process of reducing and maintaining the temperature of a body or a system below that of the surrounding atmosphere.

Refrigerator or Refrigerating system: A machine employed to reduce and maintain the temperature of a body or a system below that of the surroundings is called a refrigerator or refrigerating system.

Refrigerant: A medium which continuously extract the heat from the space within the refrigerator where temperature is to be reduced and maintained below that of the surrounding and rejects it to the surroundings.

Commonly used Refrigerant are:

- Ammonia (NH_3)
- Carbon-di-oxide (CO_2)
- Freon 12 (Dichloro-difluoro methane) and Freon 22 (Difluro-monochloro methane)
- R134A (Tetrafluoroethane)
- Sulphur dioxide (SO_2)

Properties of an Ideal or a Good Refrigerant

- Low boiling temperature
- Very low freezing temperature
- Very low specific volume
- High latent heat of vaporization
- Non-toxic
- Non-flammable and Non explosive
- Low cost
- Low specific heat of liquid and high specific heat of Vapour
- Chemically stable

Applications of Refrigeration:

- To preserve food
- To cool water
- In making ice
- To preserve drugs, blood and eye etc. in the field of medical science
- To preserve highly perishable goods
- In cryogenic application
- In air-conditioning application
- Used in dairy and form products (perfume, tooth paste, shaving scream etc)

Definitions:

1. Refrigeration effect: Refrigeration effect is the rate at which the heat is extracted by the refrigerator from the refrigerated space. Higher capacity of refrigerator means higher shall be the refrigeration effect.

Refrigeration effect is also called **capacity of refrigerator.** It is expressed in KJ/Sec.

2. Ton of refrigeration: the unit used in the field of refrigeration.

The amount of heat absorbed to produce one ton of ice in 24 hrs. when the initial temperature of water is 0°C

Usually one ton of refrigeration is taken as 210KJ/min

One ton of refrigeration = 210/60

= 3.5 KJ/sec

One ton of refrigeration = 3.5 KWatt

3. **COP:** The coefficient of performance (COP) of a refrigerator is the ratio of the energy removed from the cold reservoir to the work input. It's also known as the cooling coefficient of performance.

$$COP = \frac{Q}{W} = \frac{\text{amount of heat removed}}{\text{work input}}$$

4. Relative COP: It is the ratio of actual COP to the theoretical COP of refrigeration.

$$\boldsymbol{COP_{Relative} = \frac{COP_{Actual}}{COP_{Theoretical}}}$$

5. Ice making capacity: It is the ability of a refrigerating system to make ice starting from water at room temperature.

Principle of Refrigeration

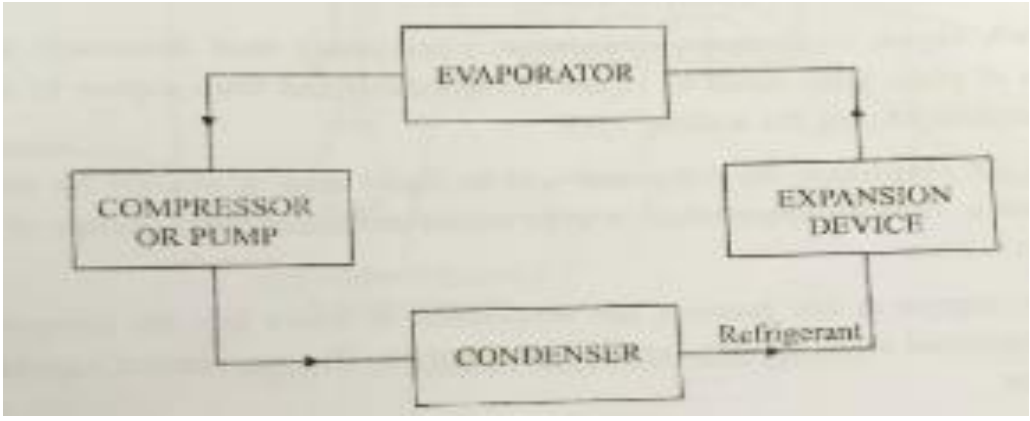


Figure: Parts of a refrigerator

The different components of VCR are:

1. Compressor or Pump
2. Condenser
3. Expansion valve
4. Evaporator

Compressor: It circulate the refrigerant to undergo refrigeration cycle. It consists of mechanical device like pump which are driven by electric motor.

Condenser: The heat from refrigerant is rejected at higher temperature to another medium, generally to atmospheric air.

Expansion valve: To reduce the pressure and temperature of the liquid refrigerant before it passed to the evaporator.

Evaporator: It is the heart of the refrigeration in which the **liquid refrigerant** is **evaporated** by **absorbing the latent heat** of **vaporization from the refrigerator** cabin in which the substance which have to be cooled are kept.

Types of Refrigeration system:

1. Vapor Compression Refrigeration (VCR) system
2. Vapor Absorption Refrigeration (VCR) system

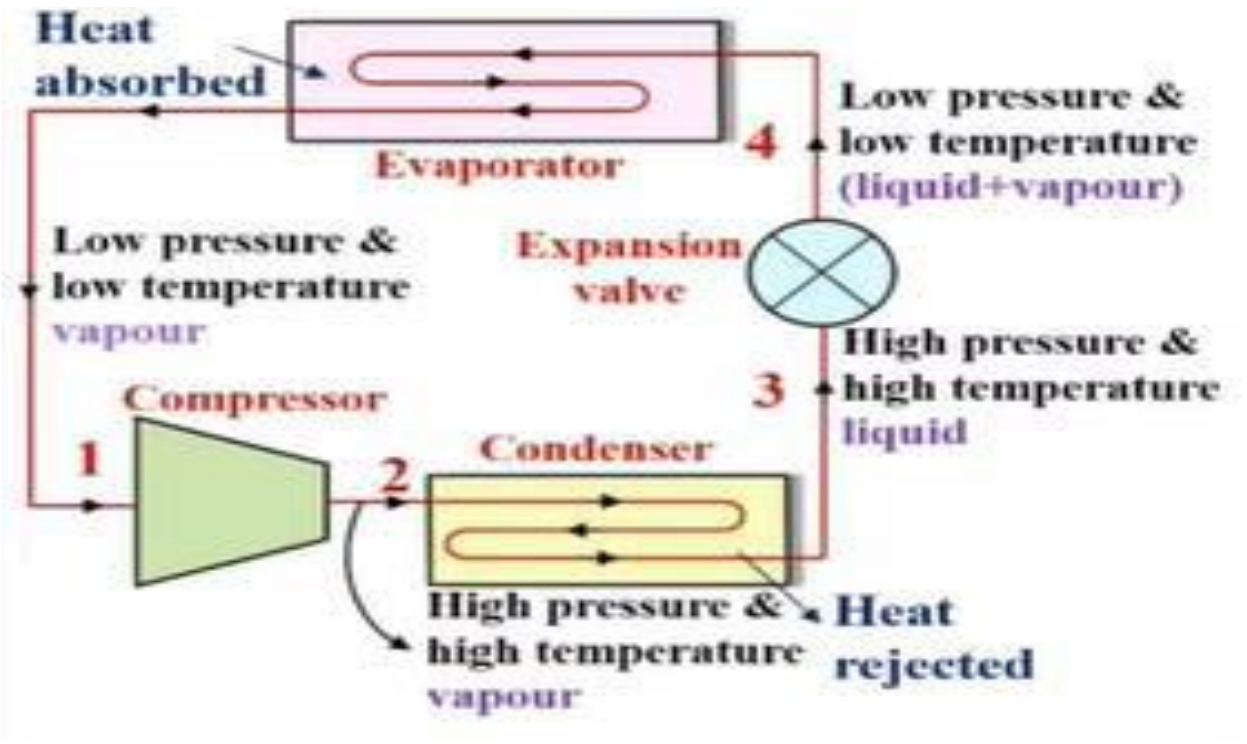


Figure: Working principle of Vapor Compression Refrigeration (VCR)

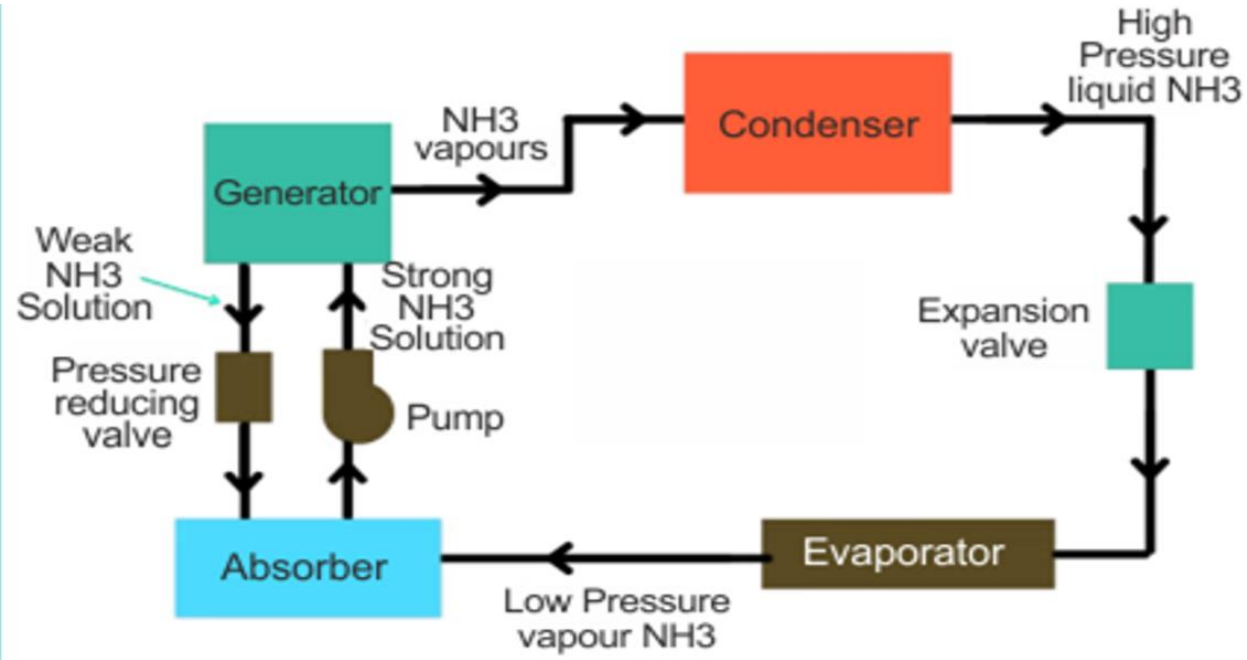
1-2: Vapour refrigerant at low pressure and low temperature is entered into the compressor where it is compressed to high pressure. So that temperature of vapour refrigerant increases. Compressor pushes vapour refrigerant to the condenser.

2-3: Condenser consists of pipe in which refrigerant gives its latent heat to the cooling medium. Therefore vapour refrigerant gets condensed in condenser. So that vapour refrigerant converts into liquid state.

3-4: In a expansion valve or throttle valve, high pressure liquid refrigerant is expanded to low pressure and low temperature.

4-1: An evaporator consists of pipe in which low pressure and low temperature liquid refrigerant is absorbed latent heat from space to be cooled. Due to absorption of heat liquid refrigerant is converted into saturated vapour. Again this vapour enters into compressor and the cycle is repeated.

2. Vapor Absorption Refrigeration (VAR) Systems



Vapour Absorption Refrigeration System

At Evaporator: The refrigerant at Low Temperature, Low Pressure Liquid Refrigerant will enter inside the evaporator coiled tubes. This refrigerant absorbs the heat from the freezing compartment and evaporates. This in turn reduces temperature of the freezing compartment. The High Temperature, Low Pressure Vapour refrigerant will enter the absorber inlet.

At Absorber: The High Temperature, Low Pressure Vapour refrigerant will enter the absorber inlet. In absorber, ammonia is mixed with water hence, it makes a high concentrated mixture. This High Temperature, Low Pressure, strong ammonia solution enter the pump.

At Pump: The High Temperature, Low Pressure, strong ammonia solution enter the pump. In the pump, the pressure. This High Temperature, High Pressure, strong ammonia solution enter heat exchanger inlet.

At Heat Exchanger: The High Temperature, High Pressure, strong ammonia solution enter heat exchanger inlet. In heat exchanger, solution is warmed by the hot weak solution flowing from heater to the absorber. This High Temperature, High Pressure, strong ammonia solution enter heater inlet.

At Heater: The High Temperature, High Pressure, strong ammonia solution enter heater inlet. In heater, solution is further heated by coil. This heating will evaporate the ammonia and separates from water thus making weak ammonia solution. Now, High Temperature, High Pressure Vapour Ammonia enters the condenser inlet.

At Condenser: The High Temperature, High Pressure Vapour Ammonia enters the condenser inlet. In condenser, vapor changes only its phase to liquid state by rejecting heat (Condensing) to atmosphere. This Low temperature, High Pressure Liquid Ammonia enters the throttling valve inlet.

At Throttling Valve: Low temperature, High pressure Liquid Ammonia enters the throttling valve inlet. In throttling valve, the pressure of the refrigerant is reduced. This reduction in pressure will ensure there is drop in boiling temperature of refrigerant. Lower boing temperature will enhance the cooling rate in the evaporator. This Low Temperature, Low Pressure Liquid Ammonia enters the inlet of evaporator.

Air conditioning:

Air conditioning is defined as the **"Process of simultaneous control of temperature, humidity, cleanliness and air-motion of the confined space"**

Working Principle of Room Air Conditioner:

Principle: An air conditioner continuously draws the air from an indoor to be cooled, cools it by the evaporator and discharges back to same indoor space that needs to be cooled. This continuous cyclic process of removing, cooling and circulation of the cooled air keeps the indoor cooler than the surrounding atmosphere.

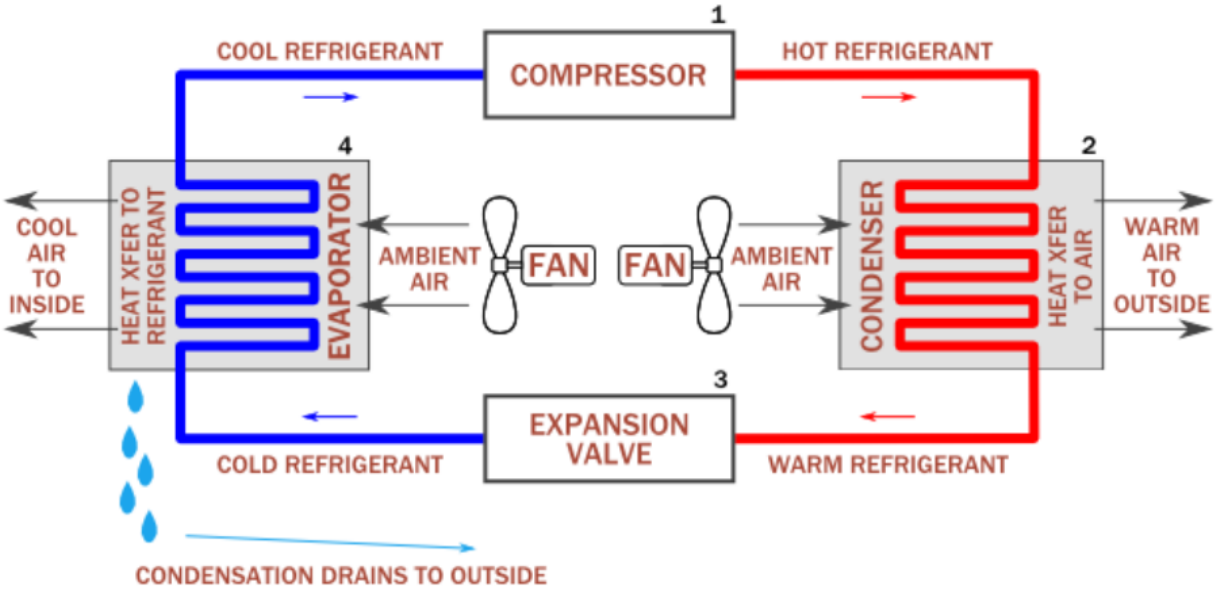


Figure: Working principle of Room Air Conditioner

Above figure shows the working principle of room air conditioner.

At Evaporator: The evaporator is **placed inside the room** to be **cooled.** The **refrigerant** from throttling valve at **Low Temperature, Low Pressure Liquid Refrigerant** will enter **inside the evaporator coiled tubes.** Behind this evaporator coil, a **blower is rotated** that draws the **hot air** from **room**. **Hot air gets cooled by passing over the evaporator coils.** Thus **cooled air is circulated into the room.** The refrigerant absorbs the heat from the hot air and evaporates. The **High Temperature, Low Pressure Vapour refrigerant** will **enter** the **compressor inlet.**

At Compressor: The refrigerant at High Temperature, Low Pressure Vapour refrigerant will enter the compressor inlet. In **compressor** is **refrigerant** is **compressed to higher pressure**. Increase in pressure will ensure there is smooth movement of refrigerant inside the tubes along with increase in temperature of refrigerant. This **High Temperature, High Pressure, Vapour refrigerant enters the condenser inlet.**

At Condenser: The **condenser is place outside the room.** The refrigerant at High Temperature, High Pressure, Vapour refrigerant enters the condenser inlet. **The hot refrigerant gets cooled** by **rejecting heat to outside atmosphere**. This **effect can be enhanced by rotating fan** behind the condenser coils. Thus formed **Low temperature, high pressure Liquid Refrigerant enters the throttling valve inlet.**

At Throttling Valve: Low temperature, High pressure Liquid refrigerant enters the throttling valve inlet. **In throttling valve**, the **pressure of the refrigerant is reduced**. This reduction in pressure will ensure there is **drop in boiling temperature of refrigerant.** Lower boing temperature will enhance the cooling rate in the evaporator. This **Low Temperature, Low Pressure Liquid Refrigerant enters the inlet of evaporator.**

Applications of Air Conditioning:

- Comfort air conditioning of auditoriums, hospitals, residences, hotels, Showrooms, dancing halls, offices etc.
- Manufacture and preservation of medicines. Surgery has found a wide application because preservation of blood and human tissues (eyes etc.) has become possible by refrigeration only.
- Storage and transportation of food stuffs such as meat, ice cream, dairy product, fish, fruit, vegetables and fruit juices etc.
- Processing of textiles, printing work and photographic material etc.
- Maintaining proper temperature in Chemical and related industries

Sl. No.	VCR	VAR
1	Work only on mechanical energy (Compressor)	Work only on heat energy
2	Vapor refrigerant is compressed	Vapor refrigerant is absorbed and heated
3	Maintenance cost is high	Maintenance cost is low
4	COP is high	COP is low
5	Operating cost is high	Operating cost is less
6	Refrigerant used is Freon - 12	Refrigerant used is Ammonia (NH_3)